

AI-Powered Enterprise E-Commerce Platform

Software Requirements Specification (SRS)

Document Information

Item	Description
Project Name	AI-Powered Enterprise E-Commerce Platform
Document Type	Software Requirements Specification
Version	1.0
Prepared By	Project Team
Status	Draft

Revision History

Version	Date	Description
1.0	DD/MM/YYYY	Initial Version

Approval Table

Role	Name	Signature
Project Supervisor		
Project Team		
Client Representative		

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete requirements of the AI-Powered Enterprise E-Commerce Platform.

The purpose of this document is to provide a detailed specification that guides:

- system design,
 - implementation,
 - testing,
 - deployment,
 - maintenance.
-

1.2 Scope

The proposed system is an enterprise-level intelligent e-commerce platform that integrates Artificial Intelligence technologies to improve customer experience and business decision-making.

The system provides:

Traditional Features

- User Management
- Authentication
- Product Management
- Inventory Management
- Shopping Cart
- Order Processing
- Payment Processing

AI Features

- Personalized Recommendations
- Fraud Detection

- Sales Forecasting
- Customer Segmentation
- AI Assistant

1.3 Objectives

The objectives of the system are:

1. Improve customer satisfaction.
2. Increase sales through recommendations.
3. Reduce fraudulent activities.
4. Improve inventory management.
5. Support business intelligence and analytics.
6. Provide an enterprise-grade software architecture.

1.4 Definitions and Acronyms

Term	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
JWT	JSON Web Token
ML	Machine Learning
RBAC	Role-Based Access Control
SRS	Software Requirements Specification
KPI	Key Performance Indicator
REST	Representational State Transfer

1.5 References

Refer to the References section.

2. Overall Description

2.1 Product Perspective

The system is a modern enterprise e-commerce platform that combines conventional online shopping features with AI-powered services.

The platform will follow a modular and service-oriented architecture.

2.2 Product Functions

The system shall provide:

Customer Functions

- Registration
- Login
- Product Search
- Product Filtering
- Shopping Cart
- Wishlist
- Order Placement
- Order Tracking

Administrative Functions

- Product Management
- User Management
- Inventory Management
- Reporting

AI Functions

- Recommendation Engine
 - Fraud Detection
 - Forecasting
 - Customer Segmentation
 - AI Chat Assistant
-

2.3 User Classes and Characteristics

Customers

Characteristics:

- Basic technical knowledge.
 - Purchase products online.
-

Administrators

Characteristics:

- Manage products and users.
 - Monitor system activities.
-

Inventory Managers

Characteristics:

- Manage stock levels.
 - Monitor demand forecasts.
-

Business Owners

Characteristics:

- Analyze reports.
 - Make strategic decisions.
-

Delivery Personnel

Characteristics:

- Update delivery information.
-

2.4 Operating Environment

Backend Environment

- Java Spring Boot
 - Python FastAPI
-

Database Environment

- PostgreSQL
 - Redis
-

Deployment Environment

- Docker
 - Linux Servers
 - Cloud Infrastructure
-

Development Environment

- VS Code
 - IntelliJ IDEA
 - Git
 - GitHub
-

2.5 Design and Implementation Constraints

Examples:

- Limited development time.
 - Learning curve for technologies.
 - Hardware limitations.
 - Dataset availability.
-

2.6 Assumptions and Dependencies

Assumptions:

1. Internet access is available.
2. External services remain operational.
3. Required datasets can be obtained.

Dependencies:

- Payment Providers
 - Email Services
 - AI Models
 - Database Services
-

3. Stakeholder Analysis

Primary Stakeholders

- Customers
 - Administrators
 - Inventory Managers
 - Business Owners
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Secondary Stakeholders

- Delivery Personnel
 - Payment Providers
 - Cloud Providers
-

Technical Stakeholders

- Developers
 - AI Engineers
 - Database Administrators
-

4. System Features

Feature 1: User Registration

Description:

Allows users to create accounts.

Priority:

High

Feature 2: Authentication

Description:

Allows secure access.

Priority:

Critical

Feature 3: Product Management

Description:

Allows management of products.

Priority:

Critical

Feature 4: Shopping Cart

Description:

Allows management of purchases.

Priority:

High

Feature 5: Order Management

Description:

Handles customer purchases.

Priority:

Critical

Feature 6: Recommendation Engine

Description:

Provides personalized recommendations.

Priority:

High

Feature 7: Fraud Detection

Description:

Detects suspicious transactions.

Priority:

Critical

Feature 8: Forecasting System

Description:

Predicts future demand.

Priority:

Medium

Feature 9: AI Assistant

Description:

Provides intelligent assistance.

Priority:

Medium

Feature 10: Analytics Dashboard

Description:

Provides business intelligence reports.

Priority:

High

5. External Interface Requirements

5.1 User Interfaces

The system shall provide:

Customer Interface

- Product pages
- Shopping cart
- Profile pages

Administrative Interface

- Product management pages
 - Analytics dashboards
-

5.2 Hardware Interfaces

The system shall support:

- Desktop Computers
 - Laptops
 - Mobile Devices
 - Tablets
-

5.3 Software Interfaces

The system shall integrate with:

- Payment APIs
- Email APIs
- AI Services
- Database Services

5.4 Communication Interfaces

Communication methods:

- HTTPS
- REST APIs
- JSON Data Exchange

6. Functional Requirements

The detailed functional requirements are described in:

04_Functional_Requirements.md

Summary:

ID Range	Description
FR-001–FR-009	User Management
FR-010–FR-017	Product Management
FR-018–FR-022	Cart and Wishlist
FR-023–FR-030	Order and Payment
FR-031–FR-033	Inventory
FR-034–FR-045	AI Services
FR-046–FR-050	Reporting

7. Non-Functional Requirements

The detailed non-functional requirements are defined in:

05_Non_Functional_Requirements.md

Categories:

- Performance
 - Security
 - Scalability
 - Reliability
 - Availability
 - Maintainability
 - Portability
 - Compliance
-

8. Business Rules

BR-001

Only authenticated users may place orders.

BR-002

Products cannot be purchased if stock is unavailable.

BR-003

Passwords shall never be stored in plain text.

BR-004

Fraud alerts shall be generated for suspicious transactions.

BR-005

Only administrators can manage products.

BR-006

Recommendation services require customer behavior data.

9. Data Requirements

Customer Data

Examples:

- Name
 - Email
 - Address
 - Purchase History
-

Product Data

Examples:

- Product Name
 - Category
 - Price
 - Inventory Information
-

Transaction Data

Examples:

- Orders
 - Payments
 - Delivery Information
-

AI Data

Examples:

- Customer Behavior
 - Product Interactions
 - Sales History
-

10. Constraints

Technical Constraints

- Java-Python integration complexity.
 - Dataset limitations.
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Resource Constraints

- Time limitations.
 - Hardware limitations.
-

Security Constraints

- Protection of customer information.
- Compliance with security standards.

11. Assumptions

1. Users possess basic digital literacy.
2. Cloud infrastructure is available.
3. External APIs remain operational.
4. Historical data can be collected.

12. Acceptance Criteria

The system will be considered successful if:

Functional Requirements

All critical requirements are implemented.

Performance Requirements

Response times meet targets.

Security Requirements

No critical vulnerabilities exist.

AI Requirements

Models achieve acceptable accuracy.

Usability Requirements

Users can perform major tasks successfully.

13. Requirement Traceability Matrix

Business Requirement	Functional Requirement	User Story	Use Case
Improve Sales	FR-034	US-014	UC-009
Reduce Fraud	FR-037	US-022	UC-010

Business Requirement	Functional Requirement	User Story	Use Case
Improve Inventory	FR-042	US-027	UC-011
Improve Decision Making	FR-048	US-032	UC-008

14. Future Enhancements

Possible future improvements:

- Mobile Application
- Multi-vendor Marketplace
- International Shipping
- Advanced Deep Learning Models
- Real-Time Recommendation System
- Voice-Based AI Assistant
- Generative AI Analytics Assistant

15. Appendices

Appendix A – Related Documents

01_Project_Vision.md

02_Stakeholders.md

03_Business_Requirements.md

04_Functional_Requirements.md

05_Non_Functional_Requirements.md

06_User_Stories.md

07_Use_Cases.md

08_Feasibility_Study.md

09_Risk_Analysis.md

Appendix B – Abbreviations

Refer Section 1.4.

16. Conclusion

This Software Requirements Specification provides a complete description of the AI-Powered Enterprise E-Commerce Platform.

This document serves as the primary reference for:

- System Architecture
- Database Design
- API Design
- Development
- Testing
- Deployment
- Maintenance

The SRS establishes a solid foundation for building an enterprise-grade intelligent e-commerce platform.

References

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- [5] Bass, L., Clements, P., & Kazman, R. (2021). Software Architecture in Practice (4th Edition).
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